

SPECIFICITY OF THE FRÖHLICH INTERACTION OF ERYTHROCYTES

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The aggregation of mixtures of red blood cells from different species of mammals provides evidence for the specificity of the Fröhlich quantum mechanical interaction between living cell membranes.

We have shown [1,2] that there is an ultra-long-range ($\sim \mu\text{m}$) attraction between human erythrocytes and that this interaction is in accord with Fröhlich's theory [3,4] of quantum-mechanical coherent vibrations in the membranes, organelles and macromolecules of living cells.

A Fröhlich interaction should be specific; that is attractive, repulsive or null, depending on the relative frequencies or phases of the vibrations in the interacting entities and on the dispersion, with frequency, of the dielectric constant of the surroundings [4]. We have adduced [2,5] the experiments of Sewchand and Canham [6] as evidence of specificity in the erythrocyte interaction. Mixtures of erythrocytes from different species will aggregate into rouleaux and Sewchand and Canham [6] showed that the ordering of the cells in the rouleaux is not random. Cells from the same species adhere to each other significantly more frequently than expected from the results of passive aggregation experiments and from theory [7]. Sewchand and Canham studied ten pairs of different species and the only exception was the combination of erythrocytes from rats and mice. Rats and mice are of the same biological family and their exception suggests that we are dealing with a living rather than a physico-chemical phenomenon [2].

The flaw in the argument for specificity is that colliding cells may come apart again. When human erythrocytes, executing brownian movement in their own plasma, touch they adhere, and, on the time scale of the slow movement, quickly slide together face to

face to form rouleaux [8]. If, in the interspecies experiments [6], this occurred with like cells but, on the contrary, unlike cells had a small probability of coming apart after contact, the results of Sewchand and Canham [6] would have a trivial explanation. Accordingly we have used our aggregation technique [2,5] on mixtures of cells. Plasma cannot be used as the suspending medium because of immunological incompatibility. The cells were suspended in phosphate-buffered-saline [5] to which had been added a rouleau-forming agent, polyvinylpyrrolidone MW 360,000 [9]. The mixture was placed in a sealed haemocytometer chamber and the aggregation process was recorded by time lapse cinémicroscopy [1,2,5,9].

We carried out four experiments with dog-human mixtures and ten experiments with cat-human mixtures, both of which are known [6] to give non-random rouleaux. In each experiment about 200 cells were in view, of which about 40 pairs would collide and 10–15 of these pairs would come from the different species. Careful scrutiny of the fourteen films revealed no case of cells, once in contact, coming apart again. The non-randomness of the arrangement of the cells in the resulting rouleaux was obvious in the cat-human experiments because of the marked difference in the diameter of the cells from these two species [6].

We also analysed the experiments by our previously described method [2,5] which measures deviations from brownian movement. Deviations are expressed as multiples of a rate constant (interaction coefficient, Ξ [2,5]). $\Xi = 1$ if the motion is brownian. $\Xi > 1$ im-

Table 1
Interaction coefficients Ξ , for explanation see text.

Cat-human mixture	Cat only or human only
4.19	6.59
3.64	5.01
4.03	5.26
3.22	7.92
6.32	6.13
	4.20
	5.20
mean 4.28 ± 1.1 SD	mean 5.76 ± 1.1 SD

plies an attraction. With a solution of polyvinylpyrrolidone rather than plasma [1,2,5] as the suspending solution, cells tend to interact variably with the surface of the haemocytometer and so can lead to inconsistent values for the drag coefficients [2,5]. This led us to reject all the dog-human experiments and five of the cat-human experiments for quantitative assessment. The interaction coefficients from the remaining five experiments were compared with those from seven experiments, four of which employed cat cells only and three of which employed human cells only (table 1). There is a difference between the two series which is highly significant by the "t" test for the differences between averages [10]. The figures suggest a reduced but positive interaction between the two species, but the variability in the figures is too great to draw firmer conclusions. These experiments will not be repeated by the current method because it has now been superseded by quasi-elastic-light scattering which is much quicker and less prone to artefact. Our published results [2,5] have been confirmed qualitatively and quantitatively by this method (O.G. Fritz Jr., private communication).

Although quantitatively these experiments were a disappointment because of technical difficulties, the observation that cells never come apart after initial

contact makes Sewchand and Canham's experiments [6] on the non-randomness of the order in mixed rouleaux into firm evidence for the specificity of the interaction between erythrocytes from different species.

Fröhlich [4] and Webb [11] have reviewed the evidence for the presence of vibrations of the predicted frequency [4] in micro organisms and in cells in tissue culture. Pohl [12,13] has detected an attraction by dividing cells for inorganic particles of high dielectric constant and we have shown that a *specific* interaction exists between erythrocytes which is dependent on a membrane potential and a supply of metabolic energy [1,2,5]. Over a limited range of organisms all the major predictions of Fröhlich's theory of coherent excitations in biological systems are confirmed.

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